# The Ideal Switch™

**Enabling the electrification of everything:**
**Milliwatts-to-kilowatts, DC-to-light.**

Oct 2020

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1879
Menlo Park, NJ
Thomas Edison patents the Circuit Breaker.

# Today, switches are everywhere.

## It's the most ubiquitous electronic component.

### In our homes.

- Managing our heat, AC, & lights.
- Our appliances are loaded with them.
- Circuit breakers that keep it all safe.

### At work.

- Control the elevators, lighting, HVAC.
- All our computers, servers, comms.
- Manage factories and all equipment inside them.

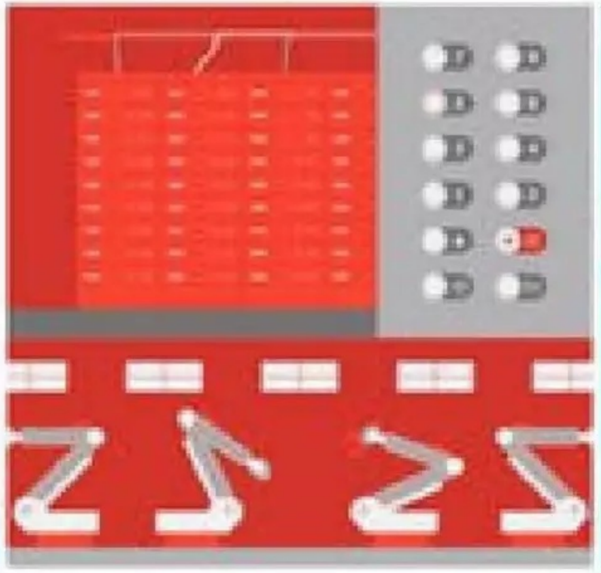
### In our cars.

- Unlock the doors and start the engine.
- Turn on and tune the radio.
- Control the wipers, windows.
- Manage the battery.

### The world at large.

- Switches control the electricity flowing from power plants to the grid to our homes.
- Switches control the data zipping around the world to our computers and smartphones.









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# Two types of switches power today's world.

## But, they're full of compromises.

### ELECTROMECHANICAL

RF



**High Power RF Coax**
High Isolation
Low Insertion Loss



**RF Surface Mount**
High Isolation
Low Insertion Loss

POWER



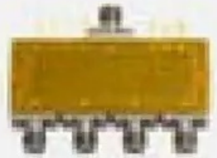
**AC/DC Power Relay**



**Surface Mount Signal Relay**

### SOLID STATE

RF



**PIN diode**
High Power
Fast Switching
Long Life



**CMOS**
Low Power
Fast Switching
Long Life

POWER



**AC/DC SSR w/heatsink**
High Voltage
High Current



**DC SSR w/heatsink**
High Current

**The Good: Can handle lots of power**

- Metal contacts: low loss and great linearity for RF signals.
- Can handle high power, eg circuit breakers or RF relays.
- "On" - doesn't generate heat under high currents.
- "Off" - air-gap open, fully isolated.

**The Bad: Big, slow, clunky and expensive**

- Moving parts limit switch rate, speed and reliability.
- Manufacturing inconsistencies (some are still assembled by hand).
- Sensitive to mechanical shock, vibration and humidity.

**The Good: Small, fast**

- Small size allows for lightweight, tighter packing.
- Inexpensive to manufacture at scale.
- Fast, silent switching.
- Immune to mechanical shock, vibration, humidity.

**The Bad: It's a semiconductor, not a conductor**

- "On" - non-zero losses, generates heat requiring large heatsinks.
- "Off" - always on, leakage currents, waste power.
- Non-linear effects of transistor distorts RF signal integrity.

# But we're still using technology from the 1830s.

And the last major step-change is from the 1970s.



**1835**
Electromechanical Relay



**1904**
Vacuum Tube



**1947**
Transistor



**1960s–70s**
Solid-State "Chip"

An entirely new "tool" for system designers to deploy.



**2016**
ideal switch



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The perfect switch:

| In theory | vs | ideal switch™ |
|---|---|---|
| When "ON" it's a wire with zero resistance. | | As low as 0.005Ω (5 Milli-Ohms). |
| When "OFF" it's an open circuit, with infinite resistance. | | Greater than 10,000,000,000Ω (10 Giga-Ohms). |
| It takes no power to turn it on and keep it on. | | Less than 0.000000000025 (25 Pico-amps). |
| It handles signals from DC to any frequency. | | DC to 50GHz+. |
| It's small, fast, silent, and lasts forever. | | Small, fast, silent, and can switch 3B+ actuations. |
| Costs $0 to make. | | Built with low-cost, scalable **semiconductor** process . . . <br>. . . but is a true metal **conductor**. |

# Menlo's "Rule of 99"

## The Ideal Switch delivers game-changing improvements to Size, Weight, Power and Cost (SWAP-C)

### Size & Weight




**>99% Reduction**

### Power Consumption




**>99% Reduction**

### Total Cost of Ownership

Other major spec improvements enable even more cost savings for system designers.

ideal switch

| Switching Speed | Lifetime | Linearity |
|---|---|---|
| **999x Faster** | **999x Longer Lasting** | **999x More Linear** |

# The Ideal Switch story.

## It starts, as most things do, with Thomas Edison.



**1879**

Edison patents the circuit breaker.

**1892**

General Electric formed, combining Edison's various companies.



Over 100 years of experience in **high voltage, high reliability** industrial controls.

**2004**

GE starts effort to reinvent the circuit breaker with a MEMS switch.



GE develops core IP, including **high temperature, high reliability** metal alloy science.

**2016**

**Menlo Micro is formed**

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# Why now? Why Menlo?

## We're a company engineered to build the Ideal Switch.

- **Why Menlo was formed**
  To scale manufacturing and drive down costs by selling into applications beyond GE's businesses.
- **Capitalized for growth**
  Over $77M raised since founding in 2016
- **Strategic partners for scaling**
  GE, Corning, Microsemi all Tier 1 players with strategic business opportunities.
- **Go-To-Market**
  Over 100K Ideal Switches shipped to Fortune 100 customer: GE Healthcare. Qualified GE supplier, ISO 2015 certified.
- **Achieved high bar for high-value medical applications**
  MRI proof-of-concept leads to Defense, Communications, Test & Measurement contracts.
- **Currently sampling from new 8-inch commercial fab, production qualification Oct 2020**
  Bringing high-power, high-reliability, low-loss switches to multiple end markets, beyond medical.

PALADIN

Microsemi

GE VENTURES

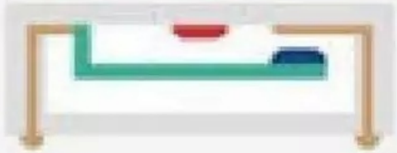
CORNING



Each GE MRI machine replaces over 100 PIN diodes with our Ideal Switch

# Introducing the Ideal Switch.

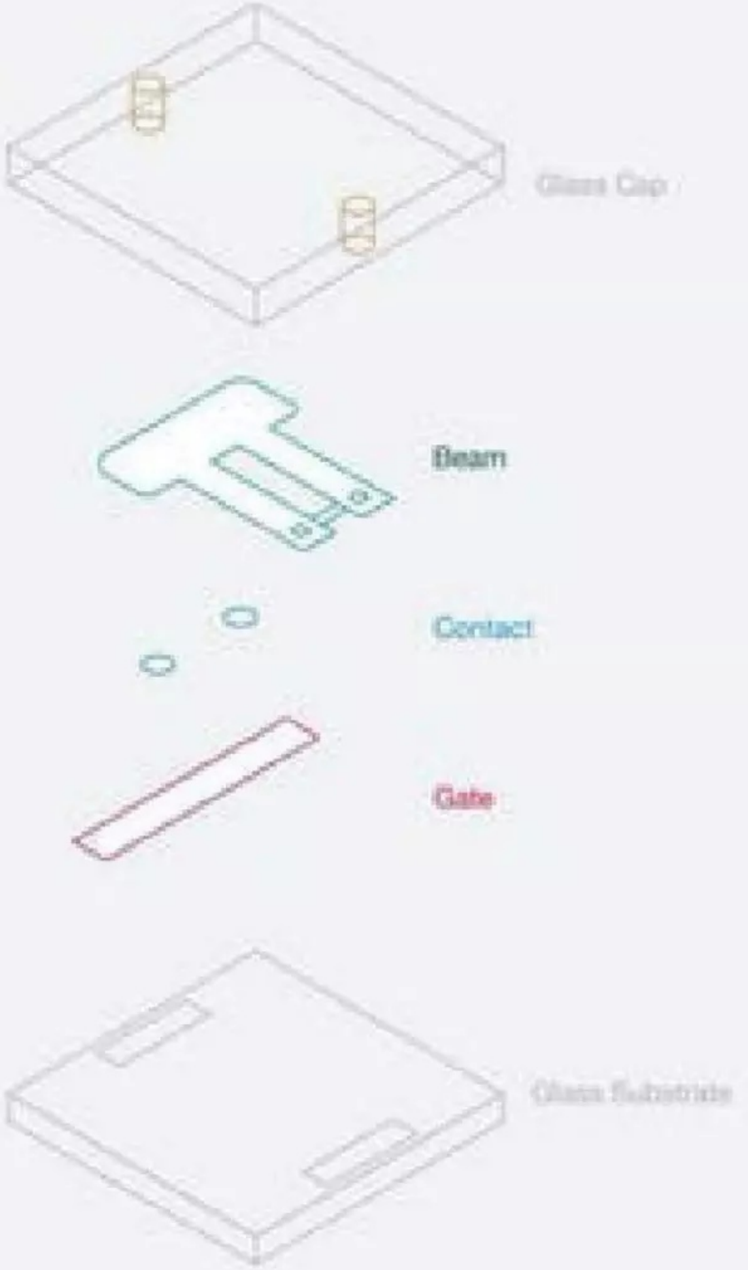
Technology platform with
breakthrough innovations
in materials and processing.



Glass Cap
Beam
Contact
Gate
Through-Glass-Via
Glass Substrate

# Introducing the Ideal Switch.

Technology platform with
breakthrough innovations
in materials and processing.




### Unique Glass Packaging

- Better thermals & better power handling, improved RF performance

### Highly Reliable

- >3B cycles spec, w/roadmap to >20B
- High-power capability

### Through-Glass-Via

- Lower parasitics, lower resistance, small-size package, lowest cost

### Simple & Scalable

- Simple design → lets us go very small (50µmx50µm unit cell)
- Small size → allows easy scaling for high power through massive switch arrays

### Scalable & Versatile

- Small size → easy to scale costs down with production volume
- Standard Process Design Kit to create many products
- Short design cycles → faster time to market



**The Packaging**
Close-up of
Through-Glass-Via

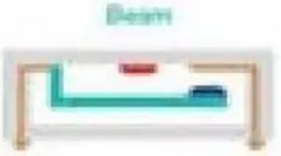
**The Beam**
Smaller than the human hair,
but can handle >400V

# Fab-Friendly advanced materials

## The secret sauce – highest reliability, superior performance, no compromises.

Menlo's beam's unique alloy goes well beyond existing ratings on:

- Power
- Temp
- Conductivity
- Reliability
- Mechanical Strength



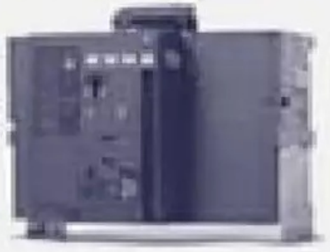

Menlo's glass substrate and TGV-enhanced glass cap deliver high:

- Conductivity
- Isolation
- Manufacturability
- Design Flexibility




### Leveraging GE's experience

Over 100 years of industrial power & controls: high voltage, high amps, high reliability.



Decades developing alloys for extreme conditions: high temp, high mechanical demands, high reliability.



### Leveraging Corning's partnership

CORNING

Over 100 years of industrial power & controls: high voltage, high amps, high reliability.



# How is it made?

## Revolutionary materials and processes combined with scalable semiconductor manufacturing techniques.

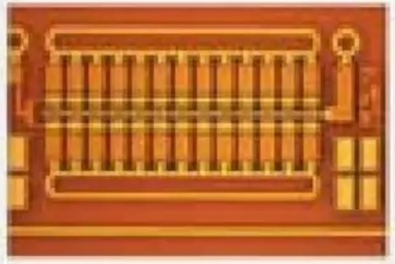
### WAFER-LEVEL MANUFACTURING



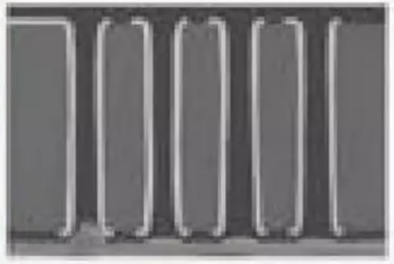
**Wafer-scale manufacturing**
on 8 in (200mm) glass substrates



**The unit cell:**
Menlo's "mechanical transistor"
- only 50µmx50µm



**Multiple arrays of unit cells**
infinite combinations to
scale up to 1000V+ and 10A+



**Through Glass Vias**
Miniaturized packaging
with highest performance



**Wafer-level**
Chip-Scale-Package
from 16mm2 to <1mm2

### APPLICATION-SPECIFIC CONFIGURATIONS



**Core Switch**
Simplest, smallest building block for multiple
power or RF applications.



**Switch with integrated controls**
Simpler control for high channel counts,
decreased need for external components.



**Multiple switch die, with routing & control**
For matrix and high-density switching
systems.



**RF System-on-Glass**
Integrate passives and other structures
directly on glass to create miniaturized RF
subsystems.

# Production capacity online

## Sampling out of commercial fab with capacity to support 5-year plan & COGS targets

Where we started:

2017



| |
|---|
| Fab capacity (Menlo's allotment) = 180 wafers/year (8" equivalent) |
| 4" wafer |
| Low volume test infrastructure = 100K units/year w/low scalability |
| Research environment = R&D pricing and T&Cs |

What have we accomplished:

Today



| |
|---|
| Fab capacity (Menlo's allotment) = 6K wafers/year |
| 8" wafer = 4x increase in die/wafer = significantly reduced COGS |
| Scalable high-volume test infrastructure = >10M units/year |
| Scalable commercial fab w/volume pricing & production quality = reduced COGS |

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# Ideal Switch vs. Electromechanical Relay.

## These improvements aren't incremental. They're paradigm shifts.

### PERFORMANCE

**100X LESS POWER**

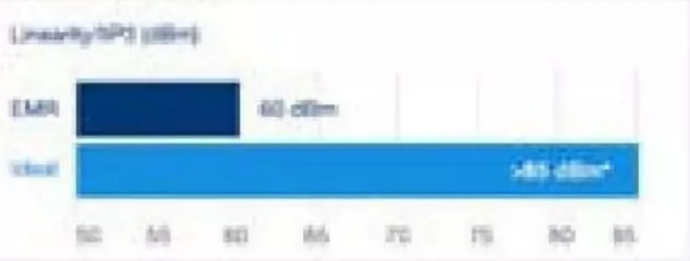
Longer battery life, less heat, new form factors




*Power needed to actuate a single switch

**100X HIGHER LINEARITY**

Distortion-free RF signal




*Ideal's RF switch compared to typical RF mass switch performance

**SWITCHES 1000X FASTER**

New capabilities. New applications.




**1000X LONGER LIFE**

Massive reliability gain




### SIZE

Conventional EMRs



71 cm³ 24 cm³ 1 cm³

ideal switch

0.6 cm³
240/30A relay with integrated protection

0.4 cm³
48/10A relay with integrated protection

.06 cm³
30V/1A signal relay

# Ideal Switch vs. Solid State.

If nano-second switching speed isn't required, Menlo's Ideal Switch is always the better option.

## PERFORMANCE

### 100X MAGNITUDE BETTER LINEARITY

More bandwidth, higher data rates




*Menlo RF switch compared to typical GaN & PIN

### >0.5DB LOWER LOSS

Longer battery life, no big, heavy heatsinks




*Menlo RF switch compared to typical GaN & PIN diode switch performance

### 10X LESS POWER

Longer battery life, less heat, new form factors




*Power needed to actuate a single "smart" switch

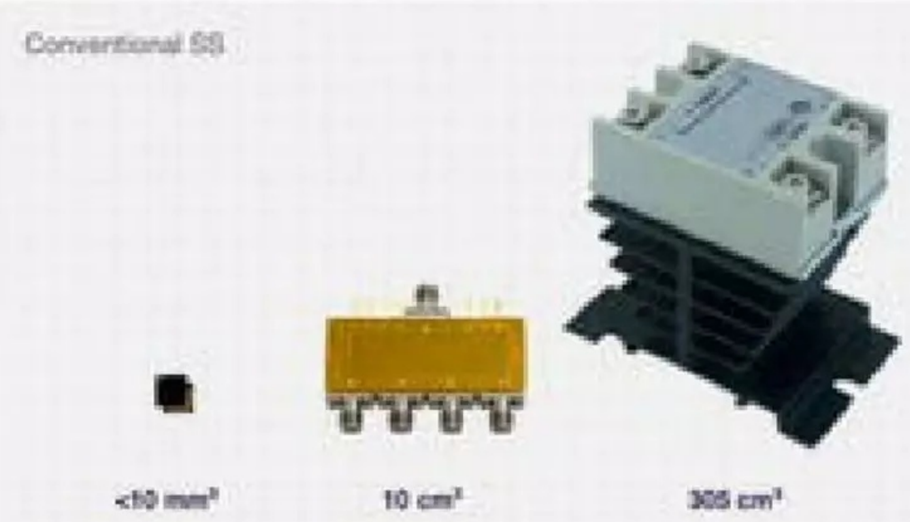
### 85% LESS RESISTANCE = NO HEATSINKS

No heatsinks = massive weight, size and cost savings




*Typical Menlo Ron for x100 contacts compared to typical x100

## SIZE




ideal switch

<10 mm²
SPST RF switch with integrated driver

0.03 cm³
High power SPST RF switch w/driver

0.6 cm³
2.5V/12A relay with integrated protection

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# The world of Ideal Switch applications

Everything will be smaller, waste less electricity, and be more cost-effective.

| | APPLICATIONS | NEEDS |
|---|---|---|
| **RF/MICROWAVE SWITCHING** | 5G cell networks<br>Radio<br>Radar<br>Satellite<br>WiFi/Connectivity | Reliability<br>Low distortion<br>Low signal losses<br>DC to mmWave frequency<br>Power efficiency |
| **GENERAL PURPOSE SWITCHING** | Telecom<br>Test & Measurement<br>Medical<br>Automotive<br>Consumer | Reliability<br>Fast switching<br>DC & Low frequency<br>Power efficiency<br>Small, light, cheap |
| **SMART AC/DC POWER SWITCHING** | Industrial IoT<br>Home automation<br>Battery management<br>Transportation<br>Motor control | Low losses, no heat<br>Speed & reliability<br>Power efficiency<br>Lightweight<br>Smart controls |



# Target Markets

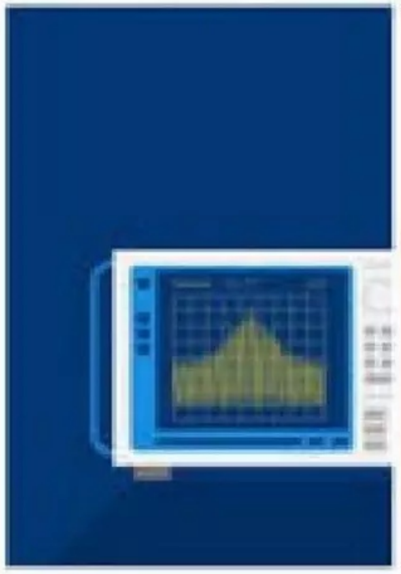
Medical



Aerospace & Defense



Test & Measurement



5G & Communications



Consumer



Industrial IoT

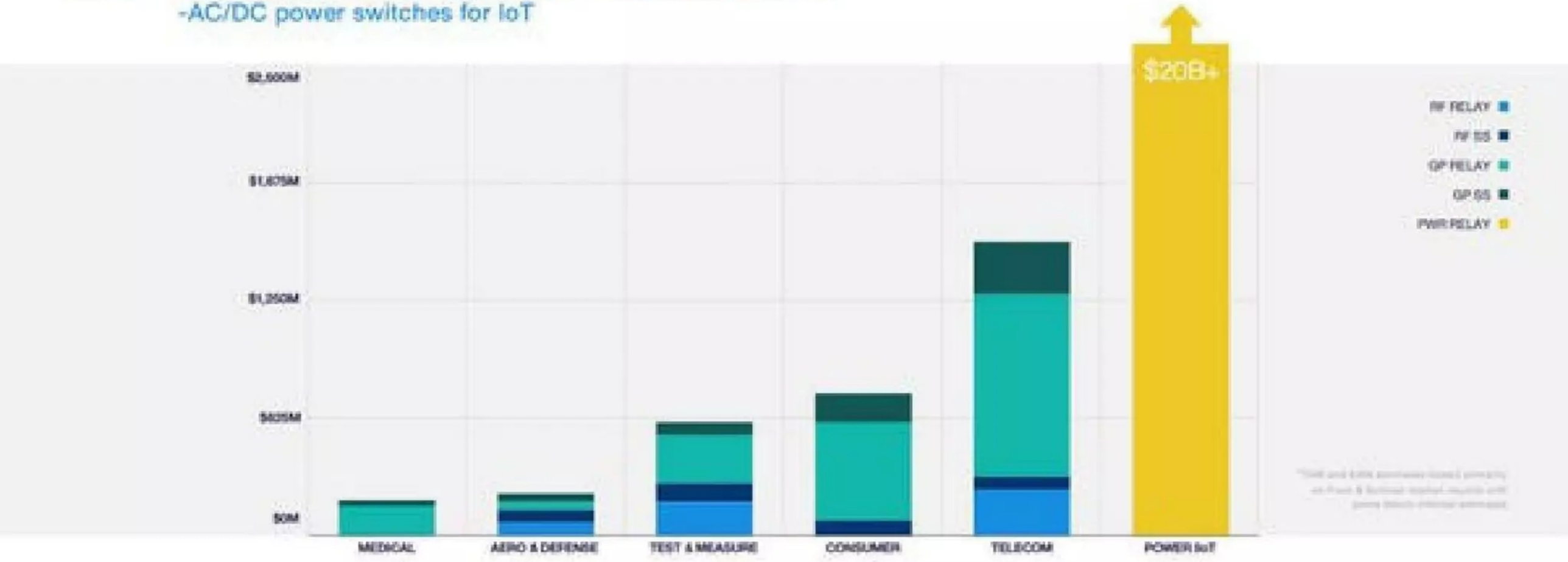


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# We are ramping to a SAM of $20B+ by 2023

Today: -RF and general purpose switching products.

Future: -High-density matrix, integrated RF subsystems, low-cost
-AC/DC power switches for IoT




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# Proven Executive Leadership Team

## Menlo Micro



**Russ Garcia**
**CEO**
EVP Microsemi, CEO WiSpry, CEO uNav, TL Silicon Systems, Hughes



**Chris Giovaniello**
**Co-Founder, SVP Marketing**
VP GE Ventures, Sales/Service/ BizDev, Teradyne



**Lew Boore**
**SVP Sales & Business Development**
VP Silanna, Cavendish Kinetics, WiSpry, HP/ Agilent, M/A-COM



**John Holtrust**
**SVP Human Resources**
Cross-functional roles and HR experience at Silicon Systems, Inc. and Intersil, Inc.



**Chris Kaimel**
**Co-Founder, CTO**
Former Senior Research Scientist at GE Global Research Center



**Jeff Baloun**
**SVP Manufacturing Operations**
GM Deca Technologies, CEO Imbera, GM Unisem Corp, CEO Atlantic Technology UK, LSI Logic



**Mark Czepiel**
**CFO**
CFO at WiSpry (RF MEMS semiconductor). Investor in INDEP



**Pete Maimone**
**SVP Product Development**
SVP of Development WiSpry, VP Product Development & Operations, uNav, Program Management, SpaceX



**Ajit Utagikar**
**VP Quality Assurance**
Zetex Technologies, Semtech Microelectronics, Analog

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# World-class Directors and Advisors

## Board of Directors



**Russ Garcia**
CEO
Menlo Micro



**Paul Conley, PhD**
Managing Director
Paladin Capital Group



**Marianne Wu**
Managing Director
40 North Ventures



**John Major**
Broadcom – Former Chairman



**Jim Cable**
pSemi – Former Chairman and CTO

## Corporate Advisors



**Rich Simoncic**
Senior Vice President, Analog Microchip Technology



**David Velasquez**
VP & GM
Corning Advanced Optics
Corning Inc.



**Kari-Pekka 'K-P' Wilska**
Nokia – Former President, America



**Michael Idelchik**
GE Global Research – Former Vice President Advanced Technology



**Tim Richardson**
Vertical Venture Partners – Partner
Micro Linear – Former CEO
ON Semi – Former Board of Directors



**Ross Stack**
RS Bioventures, LLC – Director
GE Ventures – Former Managing Director



**Steve Litchfield**
MaxLinear – CFO, Chief Strategy Officer
Microsemi – Former EVP, Chief Strategy Officer

# Thank you.